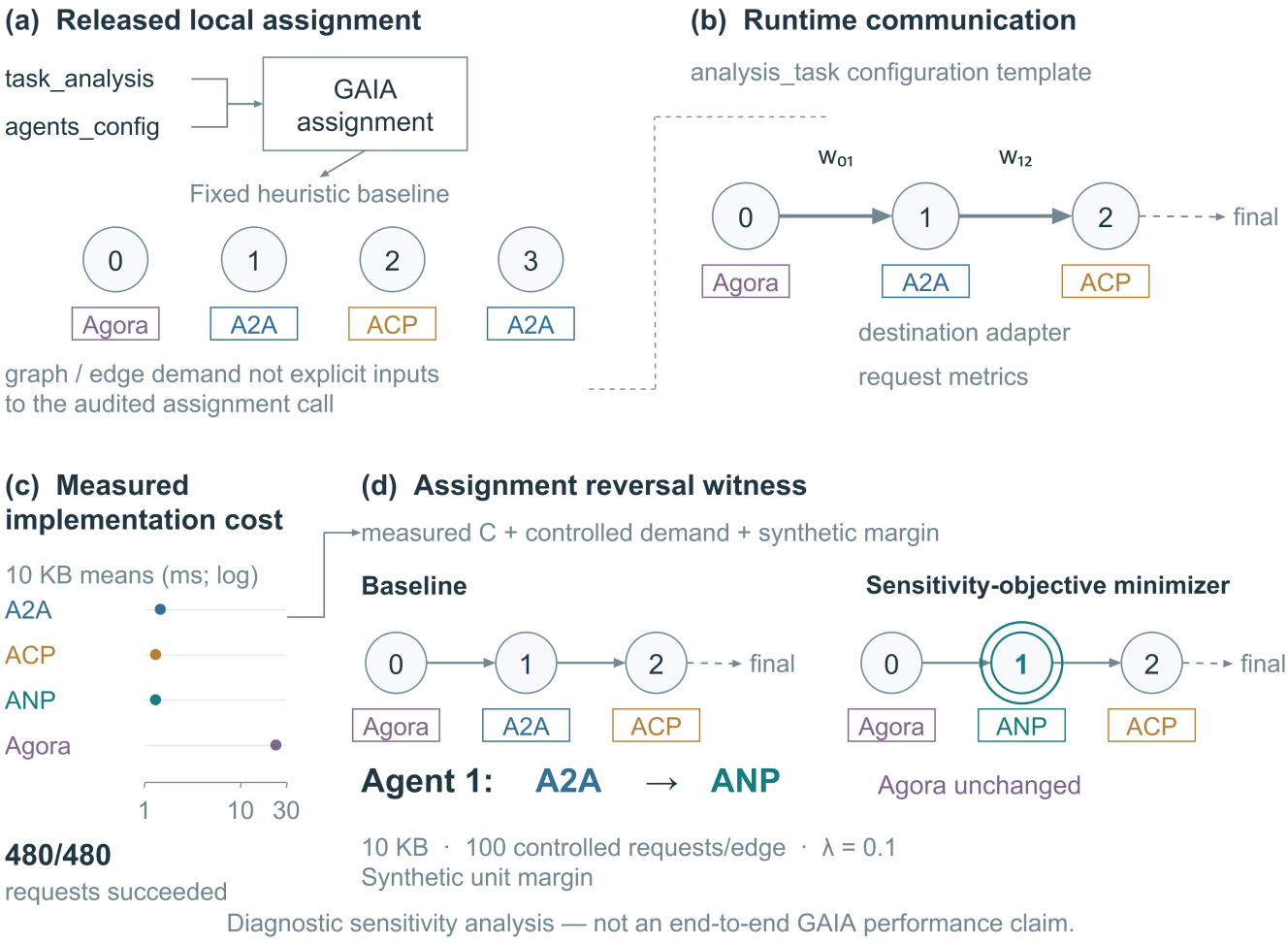
When Can Communication Cost Change a Local Protocol Assignment?

A Reproducible Diagnostic on ProtocolBench

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At commit 6349eb97, I noticed that the audited released GAIA assignment call receives task/agent inputs without explicit graph or edge-demand input; its caller uses the original heuristic fallback. Runtime communication follows destination-specific protocol paths and records request metrics. I asked whether measured local communication cost and controlled demand can change the minimizer of a declared assignment objective, with the saved baseline and synthetic margins held fixed.



The official analysis_task configuration template is instantiated under disclosed conventions; final is terminal output, not another agent. Agent 3 remains A2A outside this template's edges.

What I found

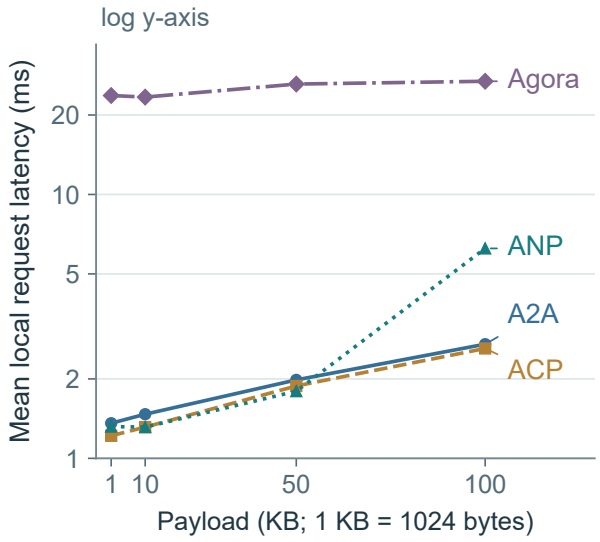
01 **Real runtime paths.** 480/480 measured requests succeeded across all four protocols.

02 **Controlled graph sensitivity.** 37/144 configurations strictly changed the declared minimizer.

03 **Repository-defined workflow.** An eligible A2A→ANP reversal persists without Agora, with forced changes separated.

Evidence and interpretation

Measured protocol latency



Local no-op request latency through released repository client/server paths. 30 measured repetitions per protocol/size after 5 warmups; setup excluded. HTTP/runtime-specific measurements; not intrinsic protocol rankings.

Timing encloses BaseAgent.send, including encode/server/no-op/decode; no LLM.
ANP's 100 KB outlier is retained; close ACP/ANP mean differences do not establish statistical superiority.

Declared sensitivity objective

$$J(\mathbf{p}) = \sum_j L_j(p_j) + \lambda \sum_{(i,j)\ \text{in}\ E} w_{ij}\, C(p_j, s_{ij})$$

L_j	synthetic local preference loss
C	measured latency, ms/request
w_{ij}	controlled request demand
λ	preference-loss units per accumulated ms

Representative witness

Workflow	analysis_task
Payload	10 KB
Demand	100 requests/edge
λ	0.1
Margin	synthetic unit loss
Change	**Agent 1: A2A→ANP**
Agora	unchanged

The exhaustive oracle finds a changed minimizer of the declared objective; this is not a GAIA task-quality claim.
Global Agora exclusion preserves this eligible reversal; forced agent-0 replacement leaves no whole-assignment gain versus the original baseline in this analysis ablation.

What this establishes — and what it does not

Supported

- Audited call: no explicit graph/edge-demand input.
- Non-identical measured local runtime costs.
- Declared objective admits reversals.
- Eligible A2A→ANP reversal without Agora.

Not established

- GAIA task-quality improvement
- Makespan improvement
- Calibrated local utility or preference margins
- Measured GAIA traffic traces
- Universal protocol rankings
- New routing-method superiority

Open question

The public release does not expose graded agent×protocol preferences or validated feasible-alternative masks for this GAIA path. If such signals are available, would it be useful to study protocol assignment jointly with communication demand under a constraint-preserving formulation?

I would be interested in exploring this direction further.

[1] Hongyi Du et al. *ProtocolBench*. ICML 2026. OpenReview.

Code: AgentProtocols @ 6349eb97 · Claim audit: claim_check_v2.md · Author: github.com/ZhijingWu.